

Usability Study on Knowledge Graph Metadata Tools (A)

Oct 29, 2025

In this study, you will complete three metadata entry tasks using different tools, upload your results, and answer short questionnaires about your experience. The session takes about 75 minutes.

* Required

Consent Form

Please read and complete the consent form before continuing:

Open the consent form: <https://forms.office.com/e/MEDWi5P3mG>

1. Did you submit the consent form? *

☐ Yes

2. Please enter your participant ID below (sent to you via email). *

Background questionnaire

3. Please rate your level of familiarity with each of the following concepts. *

	Not at all familiar	Slightly familiar	Moderately familiar	Very familiar	Extremely familiar
Semantic Web (e.g., <i>RDF</i> , <i>SPARQL</i> , <i>FAIR</i>)	☐	☐	☐	☐	☐
Metadata standards (e.g., <i>Dublin Core</i> , schema.org)	☐	☐	☐	☐	☐
Turtle syntax	☐	☐	☐	☐	☐
Data modeling (e.g., designing database schemas, JSON, XML, or ontologies)	☐	☐	☐	☐	☐
Working with structured data (e.g., tabular data, or converting unstructured data to structured data)	☐	☐	☐	☐	☐

4. How many years of experience do you have in Semantic Web technologies? *

- ☐ None
- ☐ Less than one year
- ☐ 1–3 years
- ☐ More than 3 years

Task 1: Metadata curation for the YAGO Knowledge Graph using Tool 1 (Form Interface).

Instructions

1. Access the **Knowledge graph documentation** using the following link.
This document contains metadata about the **YAGO Knowledge Graph**:
https://docs.google.com/document/d/1VXDITkl_zcl49RgoW4pEkrmI12JgYcciE/edit?usp=sharing&ouid=103705832009765615891&rtpof=true&sd=true
2. Quickly review the documentation to get a general understanding of its content.
3. Open **Tool 1 (Form Interface)** using the following link and fill out the form:
<https://anas-elghafari.github.io/knowledge-graph-metadata-form/?mode=regular>
This form displays all metadata fields related to the knowledge graph that we aim to fill in for this task. **Note ***: The form will be automatically submitted after 12 minutes, even if you have not filled in all required fields or if there are errors or warnings related to the data you entered. **Note ****: Some fields in this form are labeled as *required* or *optional*. These labels only apply to the final version of the tool, not to this study.
For this study, please ignore these labels and fill in as many fields as you can, depending on your time.
4. Submit the form.
5. Once the form has been submitted, proceed to the next section to complete the **Post-Task Questionnaire 1** about **Tool 1 (Form Interface)**.

Post-Task Questionnaire 1- Form Interface

5. How strongly do you agree or disagree with the following statements? (System Usability Scale) for Form Interface *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
I found the system unnecessarily complex.	☐	☐	☐	☐	☐
I thought the system was easy to use.	☐	☐	☐	☐	☐
I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
I found the system very awkward to use.	☐	☐	☐	☐	☐
I felt very confident using the system.	☐	☐	☐	☐	☐
I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

6. How strongly do you agree or disagree with the following statements? (Perceived Usefulness) for Form Interface

*

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
This tool enables me to accomplish metadata curation tasks.	☐	☐	☐	☐	☐
This tool supports critical aspects of my metadata curation work.	☐	☐	☐	☐	☐
Using this tool improves the quality of the metadata I curate.	☐	☐	☐	☐	☐
This tool does not support critical aspects of my metadata curation work.	☐	☐	☐	☐	☐
Using this tool increases my productivity on curating metadata.	☐	☐	☐	☐	☐

7. How strongly do you agree or disagree with the following statements? (Perceived Ease of Use) for Form Interface *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
This tool provides helpful guidance in performing tasks	☐	☐	☐	☐	☐
Interacting with this tool does not require a lot of my mental effort.	☐	☐	☐	☐	☐
Interacting with this tool is frustrating.	☐	☐	☐	☐	☐
I find it easy to get this tool to do what I want it to do.	☐	☐	☐	☐	☐

8. **How strongly do you agree or disagree with the following statements? (Intention to Use) for Form Interface** *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
I plan to use this tool again for future Knowledge Graph metadata curation.	☐	☐	☐	☐	☐
I would recommend this tool to others who work with metadata.	☐	☐	☐	☐	☐
I do not intend to use this tool again for future Knowledge Graph metadata curation.	☐	☐	☐	☐	☐

9. **What did you like most about this tool (Form Interface)?** *

10. **What difficulties did you experience while using this tool (Form Interface)?** *

11. **How could this tool (Form Interface) be improved?** *

12. **Please share any feedback about your experience using this tool (Form Interface) (positive or negative).**

Task 2: Metadata curation for the Bio2RD Knowledge Graph using Tool 2 (LLM-Assisted Form).

Instructions

1. Download the **Knowledge graph documentation** using the following link.
This document contains metadata about the **Bio2RDF Knowledge Graph**, Quickly review the documentation to get a general understanding of its content.
https://docs.google.com/document/d/1_hMJmwPKWDXiVU4593R8i8F2Rt7GddKC/edit?usp=sharing&oid=103705832009765615891&rtpof=true&sd=true
2. Open **Tool 2 (LLM-Assisted Form)** using the following link:
<https://anas-elghafari.github.io/knowledge-graph-metadata-form/?mode=llm>
This form displays the metadata fields for the knowledge graph and provides AI-generated suggestions for each field.
Note *: The form will be automatically submitted after 12 minutes, even if you have not filled in all required fields or if there are errors or warnings related to the data you entered. **Note **:** Some fields in this form are labeled as *required* or *optional*. These labels only apply to the final version of the tool, not to this study.
For this study, please ignore these labels and fill in as many fields as you can, depending on your time.
3. Upload the Knowledge graph documentation file to the tool.
4. Click on each metadata field to view the AI suggestions.
Based on the descriptions in the Bio2RDF Knowledge Graph documentation, decide whether to **accept**, **edit**, or **reject** the suggested values.
5. Once the form has been submitted, continue to the next section to fill in the **Post-Task Questionnaire 2** about **Tool 2 (LLM-Assisted Form)**.

Post-Task Questionnaire 2 -LLM-Assisted Form

13. How strongly do you agree or disagree with the following statements? (System Usability Scale) for LLM-Assisted Form *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
I found the system unnecessarily complex.	☐	☐	☐	☐	☐
I thought the system was easy to use.	☐	☐	☐	☐	☐
I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
I found the system very awkward to use.	☐	☐	☐	☐	☐
I felt very confident using the system.	☐	☐	☐	☐	☐
I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

14. How strongly do you agree or disagree with the following statements? (Perceived Usefulness) for LLM-Assisted Form

*

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
This tool enables me to accomplish metadata curation tasks.	☐	☐	☐	☐	☐
This tool supports critical aspects of my metadata curation work.	☐	☐	☐	☐	☐
Using this tool improves the quality of the metadata I curate.	☐	☐	☐	☐	☐
This tool does not support critical aspects of my metadata curation work.	☐	☐	☐	☐	☐
Using this tool increases my productivity on curating metadata.	☐	☐	☐	☐	☐

15. How strongly do you agree or disagree with the following statements? (Perceived Ease of Use) for LLM-Assisted Form

*

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
This tool provides helpful guidance in performing tasks	☐	☐	☐	☐	☐
Interacting with this tool does not require a lot of my mental effort.	☐	☐	☐	☐	☐
Interacting with this tool is frustrating.	☐	☐	☐	☐	☐
I find it easy to get this tool to do what I want it to do.	☐	☐	☐	☐	☐

16. **How strongly do you agree or disagree with the following statements? (Intention to Use) for LLM-Assisted Form ***

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
I plan to use this tool again for future Knowledge Graph metadata curation.	☐	☐	☐	☐	☐
I would recommend this tool to others who work with metadata.	☐	☐	☐	☐	☐
I do not intend to use this tool again for future Knowledge Graph metadata curation.	☐	☐	☐	☐	☐

17. **What did you like most about this tool (LLM-Assisted Form)? ***

18. **What difficulties did you experience while using this tool (LLM-Assisted Form)? ***

19. **How could this tool (LLM-Assisted Form) be improved? ***

20. **Please share any feedback about your experience using this tool (LLM-Assisted Form) (positive or negative).**

Task 3: Metadata curation for the QuoteKG Knowledge Graph using Tool 3 (Turtle Editor).

Instructions

1. Access the **Knowledge graph documentation** using the following link.
This document contains metadata about the **QuoteKG Knowledge Graph**:
https://docs.google.com/document/d/1xv6wBjulccj5HXkHJDx_IsjWMeAvoxG/edit?usp=sharing&ouid=103705832009765615891&rtpof=true&sd=true
2. Quickly review the documentation to get a general understanding of its content.
3. Open **Tool 3 (Turtle Editor)** using the following link:
<https://anas-elghafari.github.io/knowledge-graph-metadata-form/?mode=turtle>
This tool allows you to curate metadata by creating **Turtle statements**.
4. Open the **metadata specification** using the following link to see which elements should be included:
https://docs.google.com/spreadsheets/d/1pC6Hj6wl-uzYFeBmK19f_UaQyuUN4-DQ/edit?usp=sharing&ouid=103705832009765615891&rtpof=true&sd=true
Then create the corresponding Turtle statements for the metadata fields according to this specification and the QuoteKG documentation.

Note *: Some fields in this form are labeled as *required* or *optional*. These labels only apply to the final version of the tool, not to this study.
For this study, please ignore these labels and fill in as many fields as you can, depending on your time.
Note:** If you are not familiar with the Turtle syntax, you may consult this short tutorial (optional):
https://github.com/marmhan/KG-metadata-tool-usability-study/blob/main/turtle_introduction.md
Note*:** Your Turtle entry will be automatically submitted after 12 minutes, even if it is incomplete or contains validation errors.
However, if the editor is completely empty, it will not be submitted automatically. In that case, please type a single character (e.g., **x**) to enable the **Submit** button.
5. **IMPORTANT:** Once the form has been submitted, click **Export** to download your results file. It includes your results from all tasks.
Upload the exported file to the folder we shared with you by email.
6. **Once the file is uploaded,** continue to the next section to fill in the **Post-Task Questionnaire 3** about **Tool 3 (Turtle Editor)**.

Post-Task Questionnaire 3 - Turtle Editor

21. How strongly do you agree or disagree with the following statements? (System Usability Scale) for Turtle Editor *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
I think that I would like to use this system frequently.	☐	☐	☐	☐	☐
I found the system unnecessarily complex.	☐	☐	☐	☐	☐
I thought the system was easy to use.	☐	☐	☐	☐	☐
I think that I would need the support of a technical person to be able to use this system.	☐	☐	☐	☐	☐
I found the various functions in this system were well integrated.	☐	☐	☐	☐	☐
I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☐
I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☐
I found the system very awkward to use.	☐	☐	☐	☐	☐
I felt very confident using the system.	☐	☐	☐	☐	☐
I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☐

22. **How strongly do you agree or disagree with the following statements? (Perceived Usefulness) for Turtle Editor**

*

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
This tool enables me to accomplish metadata curation tasks.	☐	☐	☐	☐	☐
This tool supports critical aspects of my metadata curation work.	☐	☐	☐	☐	☐
Using this tool improves the quality of the metadata I curate.	☐	☐	☐	☐	☐
This tool does not support critical aspects of my metadata curation work.	☐	☐	☐	☐	☐
Using this tool increases my productivity on curating metadata.	☐	☐	☐	☐	☐

23. **How strongly do you agree or disagree with the following statements? (Perceived Ease of Use) for Turtle Editor**

*

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
This tool provides helpful guidance in performing tasks	☐	☐	☐	☐	☐
Interacting with this tool does not require a lot of my mental effort.	☐	☐	☐	☐	☐
Interacting with this tool is frustrating.	☐	☐	☐	☐	☐
I find it easy to get this tool to do what I want it to do.	☐	☐	☐	☐	☐

24. **How strongly do you agree or disagree with the following statements? (Intention to Use) for Turtle Editor** *

	Strongly disagree	Disagree	Neutral	Agree	Strongly Agree
I plan to use this tool again for future Knowledge Graph metadata curation.	☐	☐	☐	☐	☐
I would recommend this tool to others who work with metadata.	☐	☐	☐	☐	☐
I do not intend to use this tool again for future Knowledge Graph metadata curation.	☐	☐	☐	☐	☐

25. **What did you like most about this tool (Turtle Editor)?** *

26. **What difficulties did you experience while using this tool (Turtle Editor)?** *

27. **How could this tool (Turtle Editor) be improved?** *

28. **Please share any feedback about your experience using this tool (Turtle Editor) (positive or negative).**

Final Comparative Feedback Questionnaire

29. Which tool or tools would you prefer to use for metadata curation in the future? *

Please select at most 2 options.

- ☐ Turtle Editor
- ☐ Form Interface
- ☐ LLM-Assisted Form

30. Which tool do you think helped you complete the task most efficiently? *

- ☐ Turtle Editor
- ☐ Form Interface
- ☐ LLM-Assisted Form

31. Please share any additional comments, suggestions, or feedback about the tools.

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